MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-716/A

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the BB-716/A vented, nickel cadmium, rechargeable aircraft battery type described herein shall consist of this document and the latest issue in effect of Specification MIL-B-49450(ER).

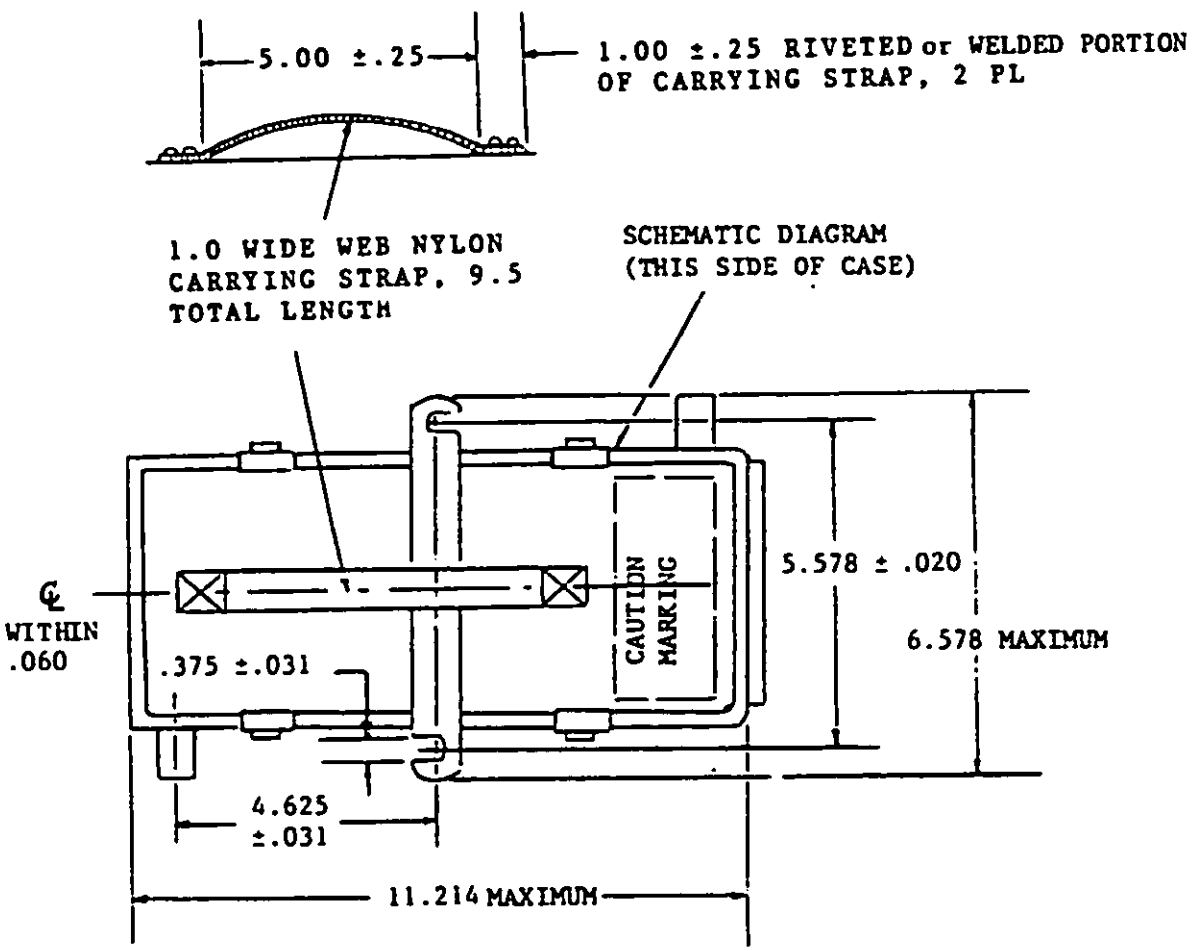
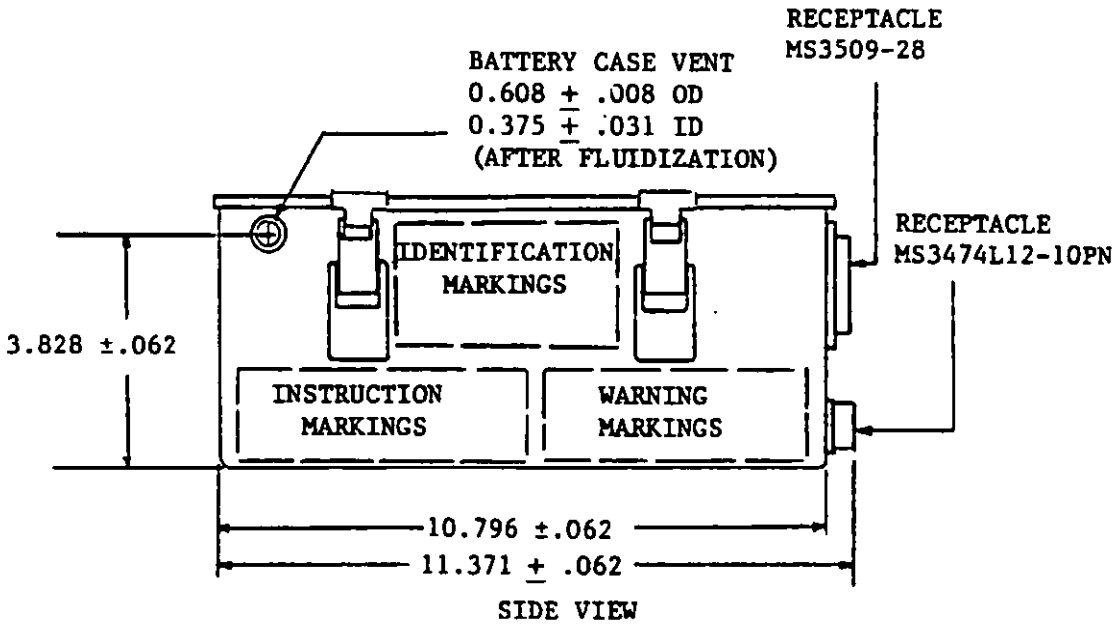


FIGURE1. MARKINGS AND DIMENSIONS OF BATTERY (TOP VIEW).

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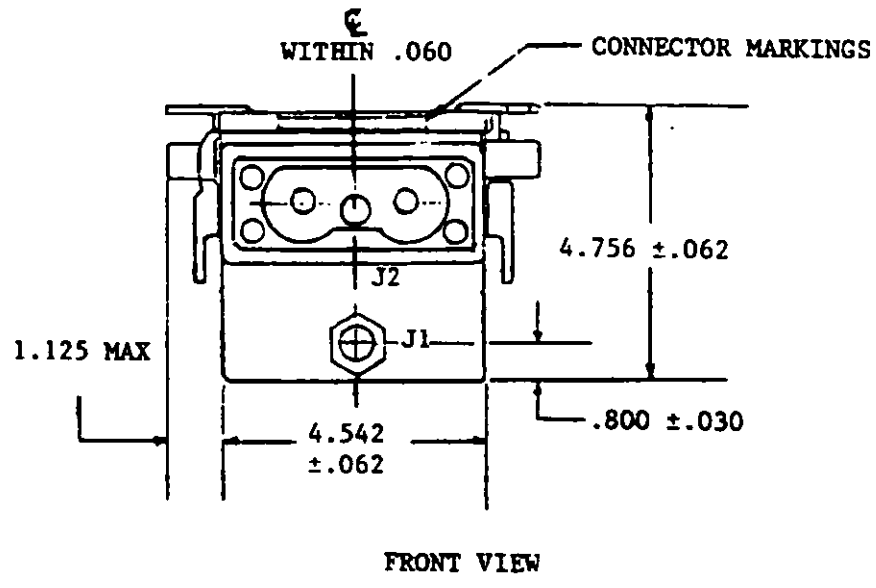


FIGURE 2. MARKINGS AND DIMENSIONS OF BATTERY (FRONT AND SIDE VIEWS).

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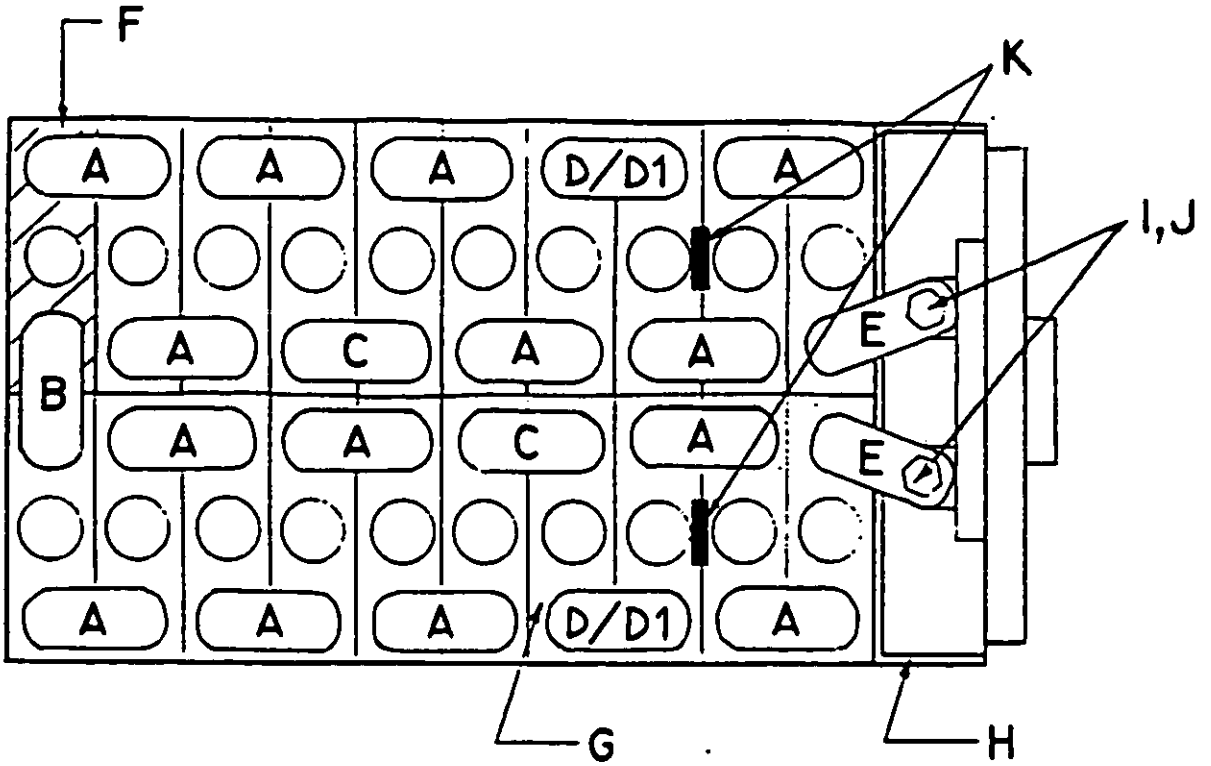


FIGURE 3. BATTERY STRAP CONFIGURATION.

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TABLE 1. REFERENCED COMPONENTS LISTING FOR FIGURE 3.

REF LTR	DESCRIPTION	PART NUMBERS	QUANTITY
A	CONNECTOR, STRAP	M49450/1-1	14
B	DIODE ASSEMBLY	M49450/2	1
C	THERMOSWITCH ASSEMBLY 1/	M49450/5-1	2
D	THERMISTOR ASSEMBLY 1,2/	M49450/5-2	2
D1	CONNECTOR, INSULATED STRAP 3/	M49450/1-3	2
E	CONNECTOR, STRAP	M49450/1-2	2
F	CELL, LOW CAPACITY	M49450/6	1
G	CELL	M49450/7	19
H	PARTITION, METAL (REMOVABLE)	--	1
I	#10-32 HEXAGONAL HEAD SCREW	AN3C3A	2
J	BELLEVILLE SPRING	M49450/3	2
K	THERMISTOR ASSEMBLY 1,3/	M49450/5-3	2

NOTES:

1. Part of Harness Assembly
2. Only applicable to Method 1 Harness Configuration
3. Only applicable to Method 2 Harness Configuration

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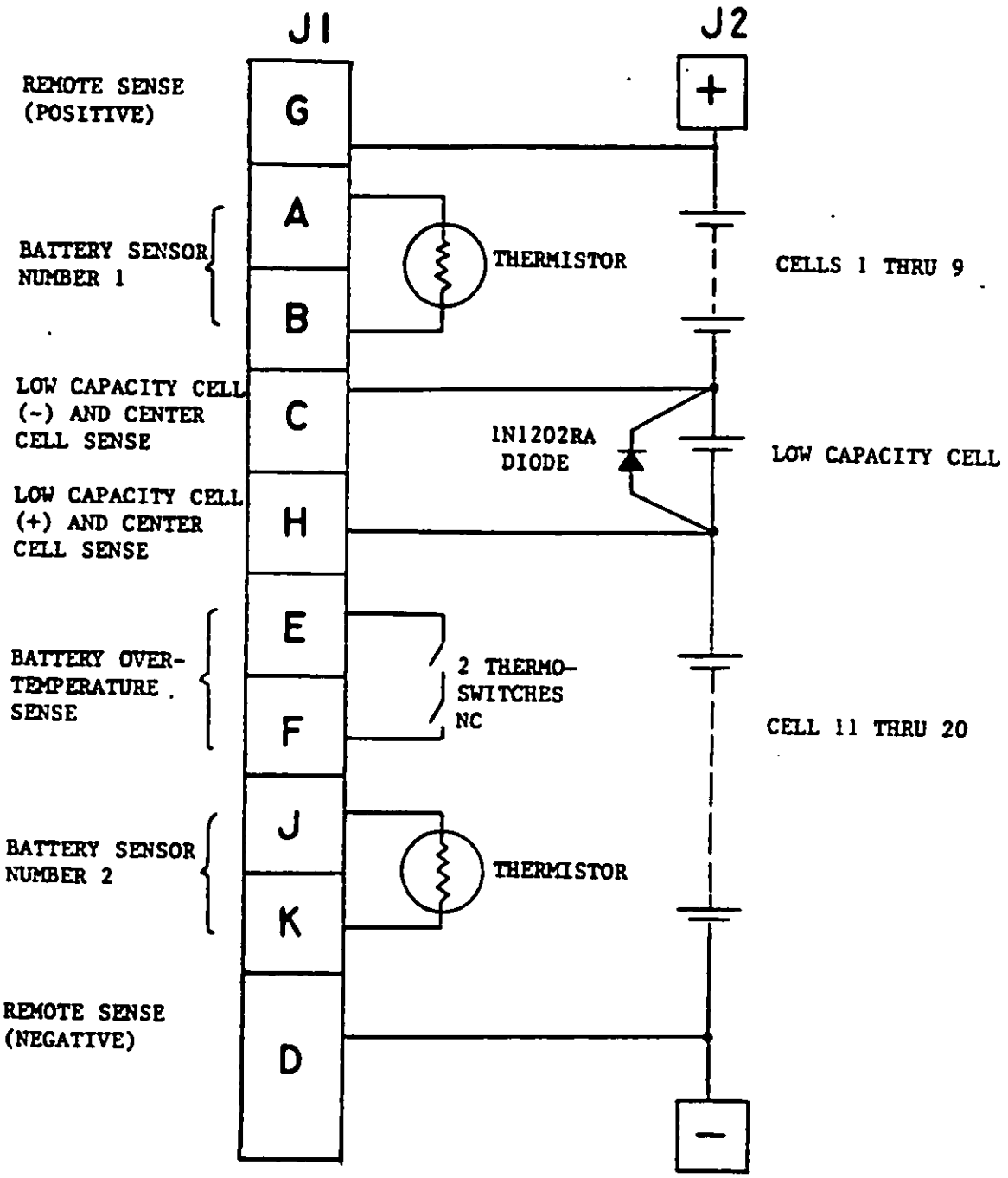


FIGURE 4. WIRING SCHEMATIC DIAGRAM FOR BATTERY.

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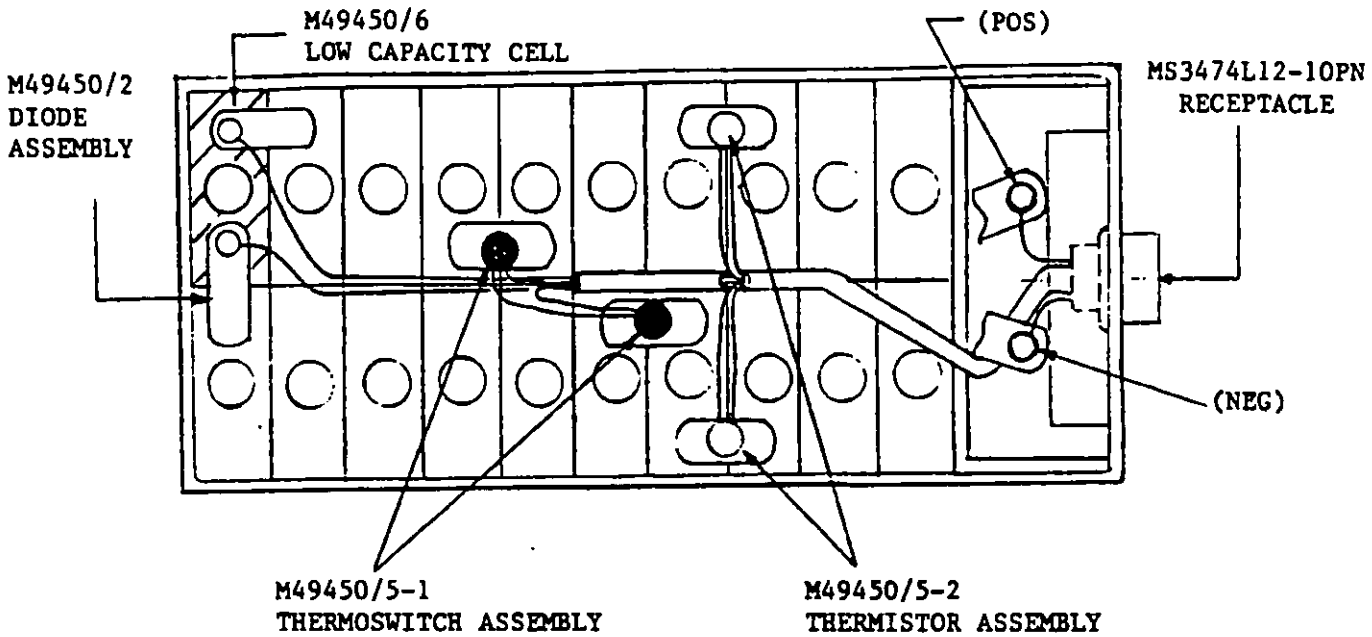


FIGURE 5. HARNESS CONFIGURATION (METHOD 1).

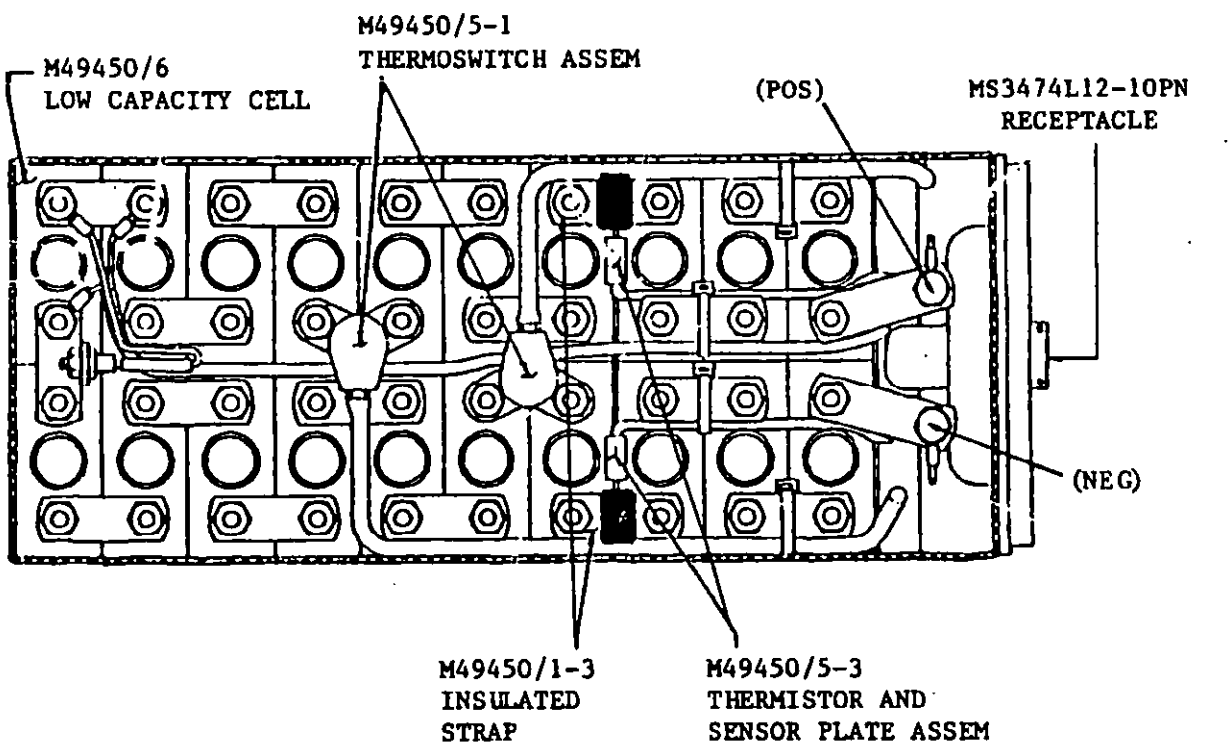


FIGURE 6. HARNESS CONFIGURATION (METHOD 2).

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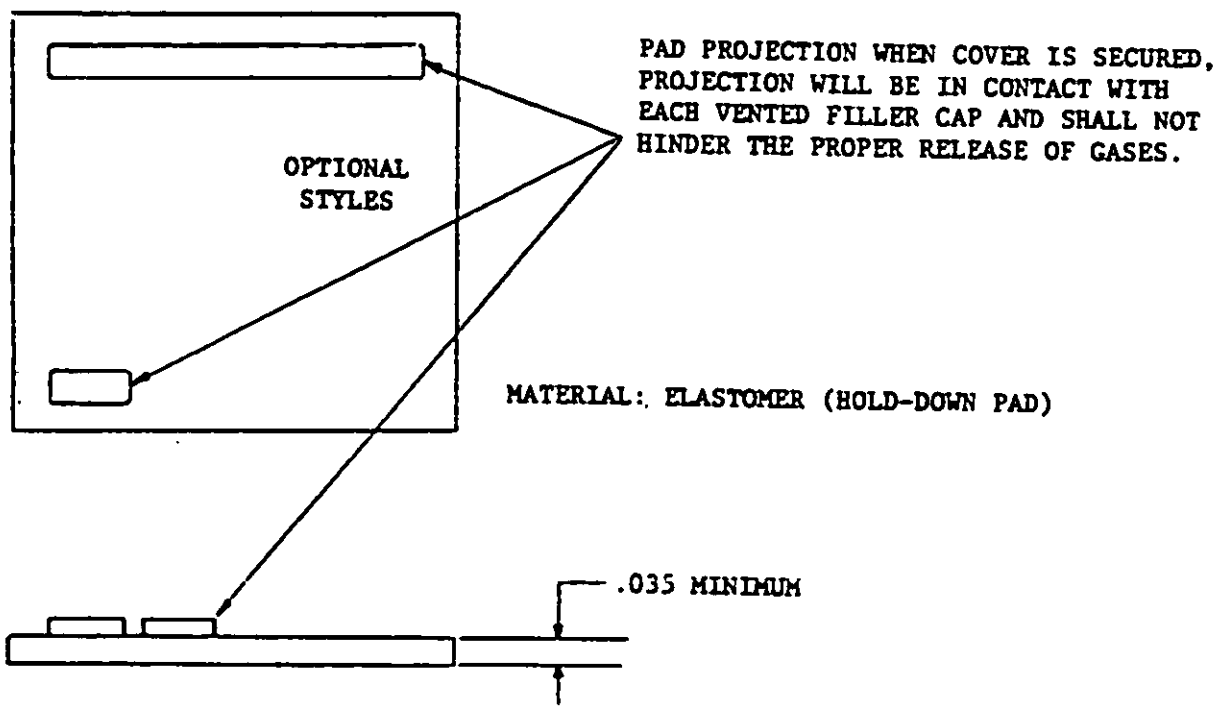


FIGURE 7. COVER HOLD-DOWN PAD.

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DRAWING NOTES:

1. All dimensions are in inches, unless specified otherwise.

REQUIREMENTS:

1. The battery's cell configuration shall conform to Figure 3 and Table 1.

2. The intercell connectors shall conform to MIL-B-49450/1(ER).

3. The diode assembly shall conform to MIL-B-49450/2(ER).

4. The harness assembly shall conform to MIL-B-49450/5(ER). The harness assembly configuration in the battery shall conform to Figure 5 or 6. Location of center cell sense/low capacity cell sense connection on either the low capacity cell terminals or adjacent terminals of equipotential is acceptable.

5. The battery shall use 19 cells conforming to MIL-B-49450/7(ER) and low capacity cell conforming to MIL-B-49450/6(ER).

6. All the battery components shall be capable of withstanding all environmental conditions to which the battery is subjected without damage or degradation of performance.

7. The wiring configuration of the battery and its components shall conform to Figure 4.

8. All dimensions in Figure 1 are prior to epoxy coating except for the following: overall length, 11.214 inches maximum; overall height, 4.756 $\pm$ .062 inch; overall width, 6.578 inches maximum.

9. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Division, American Hardware, Item Number 15834 or Nielson Latch Number 5C-B-83314-2SS or equal. Latches and strikes shall not be epoxy coated.

10. Spacers, if used, shall be Nylon per ASTM D4066-82, Class PAIII or equal; minimum thickness, .015 inch thick; color, neutral.

11. Length and width of cover pad and spacers shall conform to the inside dimensions of the applicable battery cover.

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12. The length, width and thickness of the cover projections shall provide sufficient pressure on the cell vent caps to hold down the battery cells during the Vibration Test.

13. The quantity and location of cover projection pads shall be determined by the position of the cell vents.

14. All cells within the battery shall be from the same production lot.

15. The receptacle terminal mounting block shall be 0.866 ± .015 inch from the centerline of the contact pins to the top surfaces of the terminal block. The receptacle shall be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MS3509-28.

16. Battery temperature monitoring shall be accomplished by submerging a protected thermocouple at least 5 millimeters below the surface of the electrolyte in the cell vent opening of cell number 15 (counting from the negative terminal).

17. The battery MS3474-L12-10PN receptacle shall be mounted at the battery center and at the height shown in Figure 1. The key of the receptacle shall be in the 12 o'clock position when mounted.

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REQUIREMENTS:

Nomenclature:	BB-716/A Battery, Rechargeable
Rated Capacity:	5.5 ampere-hours
Voltage (Nominal):	24 Volts
Weight (Maximum):	17.5 pounds
Connector:	one 2 pin MS3509-28 one 10 pin, circular MS3474-L12-10PN

TEST	DISCHARGE RATE (Amps)	CUT OFF VOLTAGE (V)	CAPACITY (AH) MINIMUM
Capacity	3.5	18.0	5.5
Cycle Life			
Normal Rate	5.5	18.0	5.0
High Rate	55.0	14.4	1.83 2/
Vibration	1.1	19.0	5.5
Temperature Shock	5.5	18.0	5.5
Altitude	5.5	18.0	4.6
Humidity and Charge Retention	5.5	18.0	4.1
High Rate Discharge and Low Temperature 1/	55.0	13.0	2.75
High Rate Discharge and Operating Position 1/	55.0	14.4	4.6

NOTES: 1/ By pass diode with shorting bar when low capacity cell reaches voltage of 0.5 V during high rate discharges.

2/ Discharge on cycling test limited to 2 minutes.

Preparing Activity:
Army-ER

(Project 6140-A601-08)

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